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Liver resection with and without vascular resection versus transplantation for de novo perihilar cholangiocarcinoma

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Abstract

Background and Aims: The optimal treatment strategy for de novo perihilar cholangiocarcinoma (pCCA) remains debated. This study compares outcomes between liver transplantation following neoadjuvant chemoradiation (RT+LT) and liver resection (LR), with (LR+VR) and without vascular resection (LR w/o VR).

Approach and Results: This single-center, retrospective study included patients with de novo perihilar cholangiocarcinoma treated at the Mayo Clinic Rochester (1993–2023) with curative-intent surgery. Patients underwent either (1) LR, classified as LR+VR or LR w/o VR, or (2) RT+LT following the transplant protocol. Overall survival (OS) and recurrence-free survival were analyzed using as-treated and intention-to-treat approaches, incorporating competing risk analysis and direct matching. In the as-treated analysis, RT+LT showed superior median OS compared with LR w/o VR (78.0 vs. 58.2 mo, $p = 0.03$) and LR+VR (25.8 mo, $p < 0.001$). Considering dropout rates (RT+LT: 41%, LR: 28%), intention-to-treat analysis showed no significant OS difference between LR and RT + LT (31.7 vs. 38.5 mo, $p = 0.19$). In matched as-treated analysis, RT + LT had no significant survival benefit over LR w/o VR (50.6 vs. 140.6 mo, $p = 0.08$) or LR+VR (25.8 mo, $p = 0.11$). Perioperative mortality was 4% (RT+LT), 7% (LR w/o VR), and 8% (LR + VR).

Conclusions: Both LR and RT+LT achieve excellent oncological outcomes in selected patients with de novo perihilar cholangiocarcinoma. Key challenges remain dropouts during neoadjuvant therapy in patients planned for LT and

Abbreviations: AT, as-treated; IRB, institutional review board; ITT, intention-to-treat; ISGLS, International Study Group of Liver Surgery; LR, liver resection; LT, liver transplantation; OS, overall survival; pCCA, perihilar cholangiocarcinoma; PSC, primary sclerosing cholangitis; RFS, recurrence-free survival; RT+LT, neoadjuvant chemoradiation and liver transplantation; VR, vascular resection.

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high perioperative mortality for patients undergoing LR. OS doubles in patients undergoing LR+VR compared with dropouts, supporting LR+VR as a viable option for patients who are ineligible for LT. Appropriate patient selection is crucial, as those not undergoing surgery suffer from a dismal prognosis.

Keywords: dropout rates, intention-to-treat analysis, neoadjuvant chemoradiation, survival outcomes, vascular reconstruction

INTRODUCTION

Perihilar cholangiocarcinoma (pCCA) is a highly aggressive malignancy originating in the anatomically complex hepatic hilum. It can arise in the setting of primary sclerosing cholangitis (PSC) or de novo (without the presence of PSC).[1] Surgery remains the only curative treatment, with current guidelines recommending liver transplantation (LT) for PSC-associated pCCA, complete resection for resectable de novo cases, and LT for unresectable cases that meet the transplant criteria.[2,3]

Surgical resection provides a 5-year overall survival (OS) rate of 20%–40% but presents significant technical challenges.[4–7] The tumor's proximity to major vascular structures often necessitates major hepatectomies, potentially requiring vascular resection and reconstruction, along with biliary drainage and preoperative optimization of the future liver remnant. These complexities contribute to high postoperative morbidity and mortality rates of 5%–18%.[8] In locally advanced, unresectable cases with bilateral biliary involvement (Bismuth-Corlette type IV) and/ or major vascular involvement, neoadjuvant chemoradiotherapy followed by LT has proven to be a viable option for patients meeting the transplant criteria.[9,10] This approach has demonstrated a 5-year survival rate of 50% for de novo pCCA.[11] However, neoadjuvant chemoradiation is administered at a definitive therapeutic dose, causing irreversible liver damage that inevitably leads to liver failure in the majority of patients, making subsequent LT essential and representing a point of no return.[11,12]

One of the main controversies in pCCA management arises from the absence of a standardized definition of resectability, despite ongoing efforts to reach consensus.[13] In particular, while most centers consider the need for vascular resection a criterion for unresectability and an indication for transplantation, high-expertise centers have demonstrated that liver resection (LR) with vascular reconstruction can yield acceptable outcomes.[14–20] Conversely, several studies suggest that LT should also be considered for resectable cases, given its superior outcomes compared with LR.[21,22]

In light of the diverse treatment options and the uncertainty regarding the management of de novo pCCA, this study compares 3surgical approaches: (1) neoadjuvant chemoradiation followed by LT (RT+LT), (2) liver resection without vascular resection (LR w/o VR), and (3) liver resection with vascular resection and reconstruction (LR+VR). Long-term survival outcomes across these groups were evaluated using both as-treated (AT) and intention-to-treat (ITT) approaches, accounting for patients who dropped out of treatment. To ensure a homogenous study population and minimize confounding, patients with PSC-associated pCCA were excluded, given their distinct inflammation-driven tumorigenesis and typically more favorable clinical course.[23] Conducted at a high-volume center with 30 years of multidisciplinary experience in pCCA treatment, this study aims to provide critical insights into the effectiveness and safety of each strategy, guiding optimal clinical decision-making for de novo pCCA.

METHODS

Study design and population

This single-center retrospective analysis included patients with de novo pCCA treated at Mayo Clinic Rochester between 1993 and 2023, who were intended for curative surgical treatment. Patients underwent either (1) surgical resection, beginning with exploratory laparotomy, followed by liver resection with or without vascular resection (LR+VR vs. LR w/o VR); or (2) the transplant protocol, which involved neoadjuvant chemoradiation, laparoscopic staging, and liver transplantation (RT+LT). Those enrolled in the protocol but who continued care elsewhere were excluded. This retrospective study was conducted in accordance with the principles of the Declaration of Helsinki and the Declaration of Istanbul. The study protocol was reviewed and approved by the Institutional Review Board of Mayo Clinic, Rochester, Minnesota, USA (Institutional Review Board# 22-011911). Given the retrospective nature of the study, the requirement for written informed consent was formally waived by the Institutional Review Board.

Details on institutional protocols

Management decisions were made through multidisciplinary tumor board discussions, incorporating input from hepatobiliary surgeons, transplant specialists, medical and radiation oncologists, radiologists, and pathologists.

The transplant protocol for unresectable pCCA follows strict selection criteria and treatment guidelines detailed in previous publications.[9] Patients must have a malignant-appearing stricture on cholangiography, confirmed by at least 1 of the following: positive endoscopic brushings/biopsy, CA19-9 >100 U/mL (without acute cholangitis), or a well-defined mass on imaging. Tumors must be ≤ 3 cm in radial diameter, with no intrahepatic or extrahepatic metastases. Since 2012, a positive transperitoneal biopsy has been considered a contraindication due to the associated risk of peritoneal tumor seeding. The protocol includes external beam radiation (40–45 Gy) with concurrent 5-fluorouracil, followed by brachytherapy (20–30 Gy) and maintenance capecitabine until orthotopic LT. Before transplantation, abdominal exploration confirms the absence of metastases. Dropouts were defined as cases involving medical deterioration, locoregional lymph node metastasis, or distant metastasis (including intrahepatic lesions). Transplant-listed patients receive MELD exception points per the 2002 UNOS/OPTN policy.[24]

Patients intended for curative resection underwent an imaging-based and multidisciplinary assessment to determine resectability, ensuring the possibility of an R0 resection while preserving sufficient future liver remnant. For those with confirmed vascular involvement, resection and reconstruction were tailored to the extent of tumor involvement as determined during a multidisciplinary evaluation. Vascular resections include portal vein resection, hepatic artery resection, or a combination of both. Reconstruction was performed using direct end-to-end anastomosis or graft interposition. If the future liver remnant was insufficient, preoperative portal vein embolization was conducted to enhance hepatic reserve. Patients in the LR group who, during exploratory laparoscopy, were found to have locally advanced disease or distant metastases were classified as dropouts.

Definitions

Benchmark patients for liver resection and transplantation were identified based on criteria defined in 2 prior benchmark studies.[21,25] Liver resections were classified according to the Brisbane 2000 nomenclature established by the International Hepato-Pancreato-Biliary Association, distinguishing between minor resections (removal of <3 segments) and major resections (removal of ≥ 3 segments).[26] Postoperative complications were classified according to the Clavien-Dindo classification system, with severe morbidity defined as Clavien-Dindo grade $\geq$ III.[27] Posthepatectomy liver failure was categorized based on the classification set by the International Study Group of Liver Surgery (ISGLS).[28] All specimens underwent routine histopathological workup as per American Joint Committee on Cancer (AJCC) 7th or 8th edition guidelines.[29] The decision of an R1 status was based on the microscopic tumor exposure at any biliary, vascular, or hepatic parenchymal resection margin of the surgical specimen, while R0 status was defined as a negative resection margin.

Statistical analysis

Descriptive data are presented as medians with IQRs for continuous variables and as counts with percentages for categorical variables. Continuous variables were compared using the Mann-Whitney *U* test, while categorical variables were analyzed with the chi-square test.

Survival outcomes, including OS and recurrence-free survival (RFS), were estimated using the Kaplan-Meier method, with group differences assessed by the log-rank test. OS and RFS were analyzed using both AT and ITT approaches. Time-to-event outcomes were measured from the date of surgery for the liver resection cohort and from the date of enrollment in the neoadjuvant chemoradiation protocol for the transplant cohort, until death or last follow-up without events. The number at risk was defined as individuals who had not experienced the event of interest or been censored at a given time point. To account for non–cancer-related deaths in evaluation, a competing risk analysis was performed, considering only recurrences while censoring non–cancer-related deaths.

To minimize selection bias and ensure comparable cohorts, direct matching was performed at a 1:2 ratio for the AT analysis and 1:1 for the ITT analysis. Matching was balanced based on age, sex, node-negative status, tumor size (≤ 3 cm), and CA19-9 levels.

Statistical significance was defined as a 2-sided p value of <0.05. Statistical analyses were performed using IBM SPSS Statistics version 28.0 and R Studio software (R Foundation for Statistical Computing v4.3.1, 2023-06-16).

RESULTS

Study population of liver resection and transplantation

Figure 1 presents a CONSORT diagram of this study. In the liver transplantation (RT+LT) cohort, 446 patients were initially screened for inclusion. Of these, 306 were excluded due to underlying PSC. Among the 140 eligible patients with de novo pCCA enrolled in the transplant protocol, 13 were excluded from the final

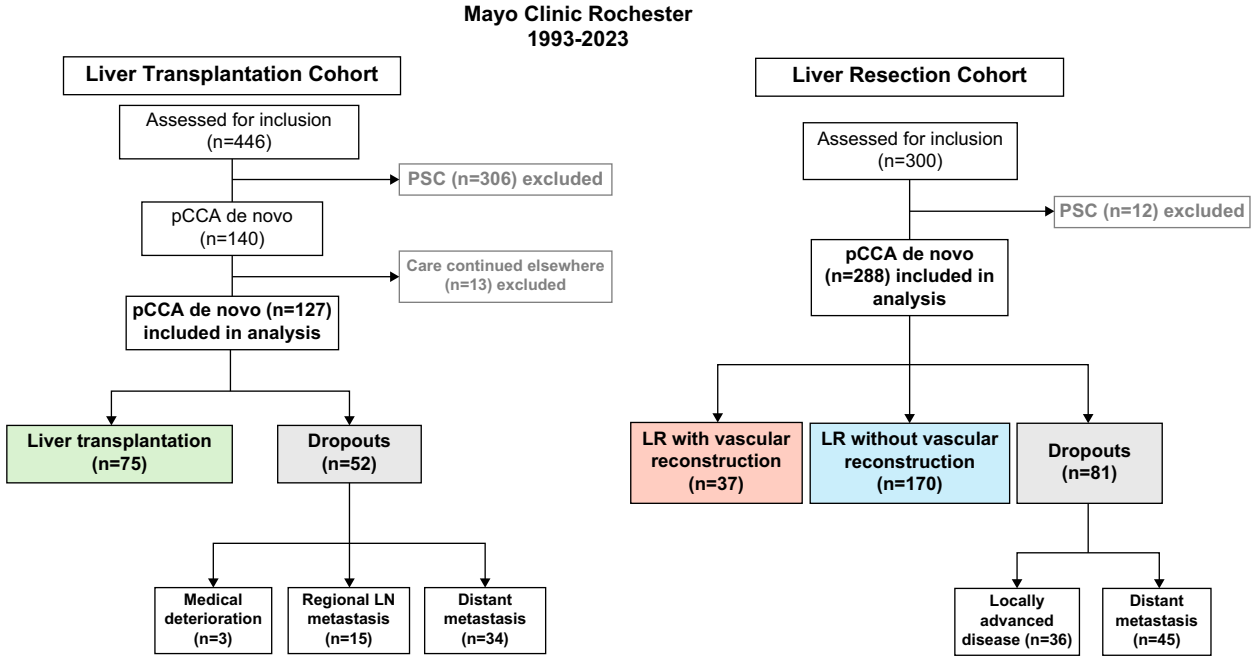


FIGURE 1 CONSORT flow diagram illustrating the workflow for determining all patients with de novo pCCA who underwent liver transplantation following neoadjuvant chemoradiation (Liver Transplantation Cohort) from 1993 to 2023, including those ultimately deemed ineligible for transplantation (referred to as "dropouts"). Similarly, all patients with de novo pCCA who received curative-intent surgery during the same period were identified through a systematic data search and were categorized into 2 groups based on the necessity of vascular resection or not (liver resection with vs. without vascular resection). Dropouts from the liver resection cohort were defined as patients who underwent exploratory laparoscopy and were found to have either locally advanced disease or distant metastasis. Abbreviations: LN, lymph node; LR, liver resection; pCCA, perihilar cholangiocarcinoma; PSC, primary sclerosing cholangitis.

analysis as they had undergone treatment at other institutions. Notably, 52 patients (41%) were classified as dropouts due to medical deterioration (n = 3), regional lymph node metastases (n = 15), or distant metastases (n = 34), resulting in 75 patients (59%), who successfully underwent LT after completing neoadjuvant chemoradiation. The primary reasons for unresectability were complex vascular involvement, precluding curative-intent surgery in 67% of cases; a low likelihood of achieving R0 resection due to extensive tumor invasion of both bile duct systems in 25%; and the presence of pCCA in the setting of underlying cirrhosis in 8%.

In the liver resection (LR) cohort, 300 patients were initially evaluated for inclusion. After excluding 12 cases with PSC diagnosed on resection specimens, 288 patients with de novo pCCA were identified. Among them, 81 patients (28%) were deemed ineligible for resection (*dropouts*) due to locally advanced disease (n = 36) or distant metastases (n = 45). As a result, 207 patients proceeded with liver resection, with 37 (13%) requiring vascular resection and reconstruction (LR+VR), while the remaining 170 (59%) underwent standard liver resection without vascular intervention (LR w/o VR). A detailed summary of the reasons for both the LR and RT+LT dropouts can be found in Supplemental Tables S1–S3, http://links.lww.com/HEP/J874.

Baseline characteristics of patients undergoing RT+LT and LR

We have summarized the comparative analyses of baseline characteristics between patients who underwent transplantation and resection (with and without vascular resection) in Table 1. Patients who underwent transplantation were generally younger than those in the LR subgroups, with a median age of 56 years compared with 62 years in the LR+VR group ($p = 0.002$) and 64 years in the LR w/o VR group ($p < 0.001$). Significant differences were also observed between the RT+LT and LR+VR, LR w/o VR groups in terms of preoperative bilirubin levels ($p = 0.028$ and $p = 0.004$), aspartate aminotransferase (AST) to platelet ratio index +albumin-bilirubin scores ($p < 0.001$ and $p < 0.001$), albumin levels ($p < 0.001$ and $p < 0.001$), and the time period of surgery ($p < 0.001$ and $p = 0.001$).

In contrast, the baseline characteristics between the LR+VR and LR w/o VR subgroups were comparable, with the only significant differences found in the frequency of preoperative portal vein embolization ($p = 0.036$) and the time period of surgery ($p < 0.001$, Table 1). Notably, within the LR+VR subgroup, portal vein reconstruction was performed in 84% of cases, isolated hepatic artery reconstruction in 11%, and combined reconstruction in 5%.

TABLE 1 Baseline characteristics of all patients with de novo pCCA with versus without vascular resection (LR+VR, LR w/o VR) and those following neoadjuvant chemoradiation and liver transplantation (RT+LT)

	Baseline characteristics of LR+VR vs. LR w/o RT vs. RT+LT					
Variable	**LR+VR (N = 37)**	**LR w/o VR (N = 170)**	**RT+LT (N = 75)**	***p* LR+VR vs. RT+LT**	***p* LR w/o VR vs. RT+LT**	***p* LR+VR vs. LR w/o VR**
Sex, n (%)				0.077	0.095	0.526
Male	21 (56.8)	106 (62.4)	55 (73.3)			
Female	16 (43.2)	64 (37.6)	20 (26.7)			
Age (y)	62 (49.5–74.5)	66.0 (56.0–72.0)	56.0 (48.1–62.0)	0.002	< 0.001	0.627
BMI (kg/m^2)	25.3 (22.3–30.1)	25.6 (22.7–29.6)	25.7 (22.8–29.3)	0.975	0.796	0.758
ECOG, n (%)				0.807	0.278	0.689
0	26 (70.3)	127 (74.7)	51 (68.0)			
> 0	11 (29.7)	43 (25.3)	24 (32.0)			
Bilirubin (mg/dL)	1.0 (0.5–2.8)	1.1 (0.6–2.9)	1.4 (1.0–2.1)	0.028	0.004	0.355
APRI+ALBI	−2.06 (−2.49 to −1.51)	−2.03 (−2.46 to −1.44)	−2.61 (−3.0 to −2.0)	< 0.001	< 0.001	0.927
Prothrombin time (s)	11.6 (11.1–12.4)	11.4 (10.5–12.3)	11.0 (9.5–12.6)	0.122	0.152	0.462
Platelets (×10^9/L)	240 (188–318)	255 (203–325)	239 (170–319)	0.920	0.153	0.288
INR	1.0 (1.0–1.1)	1.0 (1.0–1.1)	1.0 (1.0–1.1)	0.411	0.298	0.954
AST (U/L)	57.0 (37.5–81.0)	57.0 (39.0–85.0)	60.5 (33.3–103.8)	0.802	0.915	0.822
ALT (U/L)	70.0 (44.0–120.0)	69.0 (45.0–119.0)	64.0 (39.3–112.8)	0.681	0.449	0.978
Albumin (g/dL)	4.1 (3.8–4.3)	4.0 (3.7–4.2)	3.9 (3.7–4.1)	< 0.001	< 0.001	0.297
CA19-9 (U/mL)	76 (35–386)	76 (26–344)	143 (52–514)	0.527	0.055	0.661
Bismuth type, n (%)				NA	NA	< 0.001
Type 1	2 (5.4)	3 (1.8)				
Type 2	4 (10.8)	7 (4.1)				
Type 3a	19 (51.4)	69 (40.6)				
Type 3b	8 (21.6)	79 (46.5)				
Type 4	4 (10.8)	12 (7.0)				
Reasons for unresectability, n (%)				NA	NA	NA
Complex vascular involvement			50 (67)			
Low likelihood of R0			19 (25)			
Cirrhosis			6 (8)			
Type of resection, n (%)			NA	NA	NA	0.100
Right hepatectomy	16 (43.3)	49 (28.8)				
Extended right hepatectomy	9 (24.3)	29 (17.1)				
Left hepatectomy	7 (18.9)	69 (40.6)				
Extended left hepatectomy	5 (13.5)	19 (11.2)				
Central hepatectomy	0 (0)	4 (2.3)				
PVE, n (%)	8 (21.6)	16 (9.4)	NA	NA	NA	0.036
Time period of surgery, n (%)				< 0.001	0.001	< 0.001
before 2000	2 (5.4)	0 (0.0)	4 (5.3)			
2000–2010	4 (10.8)	52 (30.6)	32 (42.7)			
2011–2020	18 (48.7)	90 (52.9)	34 (45.3)			
2021–2024	13 (35.1)	28 (16.5)	5 (6.7)			

TABLE 1. (*continued*)

Variable	Baseline characteristics of LR+VR vs. LR w/o RT vs. RT+LT					
	LR+VR (N = 37)	LR w/o VR (N = 170)	RT+LT (N = 75)	*p* LR+VR vs. RT+LT	*p* LR w/o VR vs. RT+LT	*p* LR+VR vs. LR w/o VR
Type of vascular reconstruction, n (%)		NA	NA	NA	NA	NA
Portal vein (PV)	31 (83.8)					
Hepatic artery (HA)	4 (10.8)					
Combined	2 (5.4)					
Benchmark criteria, n (%)						
Yes	NA	72 (42.4)	54 (72.0)	NA	< 0.001	NA

The data presented are summarized as median values with corresponding IQRs.
Abbreviations: ALBI, albumin-bilirubin score; APRI, aspartate aminotransferase (AST) to platelet ratio index; BMI, body mass index; CA19-9, carbohydrate antigen 19-9; ECOG PS, Eastern Cooperative Oncology Group performance status; INR, international normalized ratio; LR w/o, VR liver resection without vascular resection; LR+VR, liver resection with vascular resection; NA, not available; pCCA, perihilar cholangiocarcinoma; PVE, portal vein embolization; RT+LT, neoadjuvant chemoradiation and liver transplantation.

Comparison of perioperative outcomes and clinicopathological parameters

Patients with LR+VR showed significantly higher incidence of R1 resections, Clavien-Dindo grade ≥ III postoperative complications, and differed in terms of adjuvant therapy as compared with LR w/o VR (Table 2).

In the RT+LT cohort, tumors were significantly smaller, with a median size of 1.7 cm, compared with the LR subgroups (3.0 cm in LR+VR, $p < 0.001$; 2.8 cm in LR w/o VR, $p < 0.001$). Tumor grading also showed significant variation between the RT+LT and LR subgroups, with 38.7% of transplanted patients exhibiting a complete response on final pathology.

OS in AT and ITT analyses for all-comers and after matching

Median follow-up times were 16.3 years (95% CI: 11.3–20.5) for RT+LT, 5.8 years (95% CI: 2.2–NA) for LR+VR, and 7.0 years (95% CI: 5.0–8.4) for LR w/o VR. RT+LT was associated with significantly better median OS compared with LR w/o VR (78.0 mo [95% CI: 56.6–113.0] vs. 58.2 mo [95% CI: 51.6–82.1], $p = 0.03$) and LR+VR (25.8 mo [95% CI: 19.7–82.6], $p < 0.001$, Figure 2A). The 5-year OS rates were 58.6% for RT+LT, 49.6% for LR w/o VR, and 26.7% for LR+VR. In the ITT analysis, however, no significant difference in median OS was observed between LR and RT+LT (31.7 mo [95% CI: 25.8–39.2] vs. 38.5 mo [95% CI: 22.9–56.4], $p = 0.19$).

In the AT analysis of matched cohorts (RT+LT, n = 32; LR w/o VR, n = 32; LR+VR, n = 16), the RT+LT cohort no longer demonstrated a statistically significant survival benefit compared with LR+VR (50.6 mo, 95% CI: 39.2–97.2 vs. 25.8 mo, 95% CI: 18.4–NA, $p = 0.11$) or LR w/o VR (140.6 mo, 95% CI: 52.9–NA, $p = 0.08$). Among the matched cohorts, the 5-year OS rate was the highest in LR w/o VR (60.8%), followed by RT+LT (44.2%) and LR+VR (23.6%). Similarly, the matched ITT analysis (RT+LT, n = 51; LR, n = 51) revealed no significant difference in median OS between RT+LT and LR ($p = 0.8$). A detailed summary of the matched cohorts for both AT and ITT analyses is provided in Supplemental Tables S4 and S5, http://links.lww.com/HEP/J874.

A comparative OS analysis was then conducted for the RT+LT, LR+VR, and LR w/o VR groups, incorporating the respective dropout cohorts for RT+LT and LR (Figure 3). The results revealed significantly worse median OS for RT+LT dropouts (13.0 mo), with similarly poor survival outcomes in the LR dropout cohort (14.7 mo).

When diagnostic laparoscopy was considered a staging procedure rather than an indicator of treatment intent, patients with distant metastasis (eg, peritoneal carcinomatosis) were excluded. This modification did not impact OS outcomes, which remained comparable between RT+LT and LR in both the all-comers cohort ($p = 0.89$) and after matching ($p = 0.17$), as illustrated in Supplemental Figures S1 and S2, http://links.lww.com/HEP/J874.

RFS and cumulative incidence analysis incorporating competing risks for all-comers and after matching

In the all-comers cohort, RFS analysis initially identified RT+LT as the most favorable group (median RFS: 46.7 mo), followed by LR w/o VR (median RFS: 32.3 mo), with LR+VR demonstrating the lowest RFS (median RFS: 17.7 mo). Pairwise comparisons confirmed significant differences between RT+LT versus LR w/o VR ($p = 0.01$), RT+LT versus LR+VR ($p = 0.001$), and LR w/o VR versus LR+VR ($p = 0.02$, Figure 4A).

TABLE 2 Comparison of perioperative outcomes and clinicopathological parameters between patients with pCCA who underwent resection with (LR+VR) versus without vascular resection (LR w/o VR) and patients with pCCA who underwent transplantation (RT+LT)

Perioperative outcomes and clinicopathological parameters of LR+VR vs. LR w/o RT vs. RT+LT						
Variable	**LR+VR (N = 37)**	**LR w/o VR (N = 170)**	**RT+LT (N = 75)**	***p*** **LR+VR vs. RT+LT**	***p*** **LR w/o VR vs. RT+LT**	***p*** **LR+VR vs. LR w/o VR**
Resection margin, n (%)				NA	NA	0.033
R0	25 (67.6)	141 (82.9)				
R1	12 (32.4)	29 (17.1)				
Hepatic parenchymal	5 (13.5)	10 (5.9)				
Vascular	5 (13.5)	3 (1.8)				
Bile duct	8 (21.6)	21 (12.3)				
PHLF grade B/C, n (%)	10 (27.0)	27 (15.9)	NA	NA	NA	0.109
Morbidity, n (%)				0.377	0.178	0.043
Clavien-Dindo 0	4 (10.8)	46 (27.0)	17 (22.7)			
Clavien-Dindo I	3 (8.1)	26 (15.3)	7 (9.3)			
Clavien-Dindo II	9 (24.3)	36 (21.2)	18 (24.0)			
Clavien-Dindo III	11 (29.8)	31 (18.2)	19 (25.3)			
Clavien-Dindo IV	5 (13.5)	19 (11.2)	11 (14.7)			
Clavien-Dindo V	3 (8.1)	12 (7.1)	3 (4.0)			
NA	2 (5.4)	0 (0.0)	0 (0.0)			
90 d mortality, n (%)	3 (8.1)	12 (7.1)	3 (4.0)	0.364	0.357	0.823
Pathologic stage (AJCC 8th edition), n (%)			NA	NA	NA	0.346
Stage 0	1 (2.7)	0 (0)				
Stage 1	4 (10.8)	22 (13.0)				
Stage 2	16 (43.2)	52 (30.6)				
Stage 3	13 (35.2)	90 (52.9)				
Stage 4	3 (8.1)	6 (3.5)				
Tumor size (cm)	3.0 (2.7–3.9)	2.8 (2.2–3.7)	1.7 (1.2–2.0)	< 0.001	< 0.001	0.061
LN status				NA	NA	0.542
N0	19 (51.4)	83 (48.8)				
N1	18 (48.6)	76 (44.7)				
N2	0 (0.0)	3 (1.8)				
Nx	0 (0.0)	8 (4.7)				
Number of examined LNs	8 (4–13)	8 (4–12)		NA	NA	0.612
Histologic grade, n (%)				< 0.001	< 0.001	0.212
G0	1 (2.7)	0 (0)	29 (38.7)			
G1	7 (18.9)	17 (10.0)	5 (6.7)			
G2	15 (40.6)	82 (48.2)	25 (33.3)			
G3	14 (37.8)	65 (38.2)	16 (21.3)			
G4	0 (0.0)	6 (3.6)	0 (0)			

The data presented are summarized as median values with corresponding IQRs).
Abbreviations: AJCC, American Joint Committee on Cancer; LNs, lymph nodes; LR, liver resection; NA, not available; PHLF, posthepatectomy liver failure; RT + LT, neoadjuvant chemoradiation and liver transplantation; VR, Vascular resection.

To refine the analysis, we performed a cumulative incidence assessment of recurrence while accounting for non–cancer-related death as a competing risk. In the all-comers cohort, LR w/o VR had the lowest cumulative incidence of recurrence, followed by RT+LT, while LR +VR remained the least favorable. Statistical comparisons revealed no significant difference between LR w/o VR and RT+LT ($p = 0.31$), whereas LR w/o VR had a

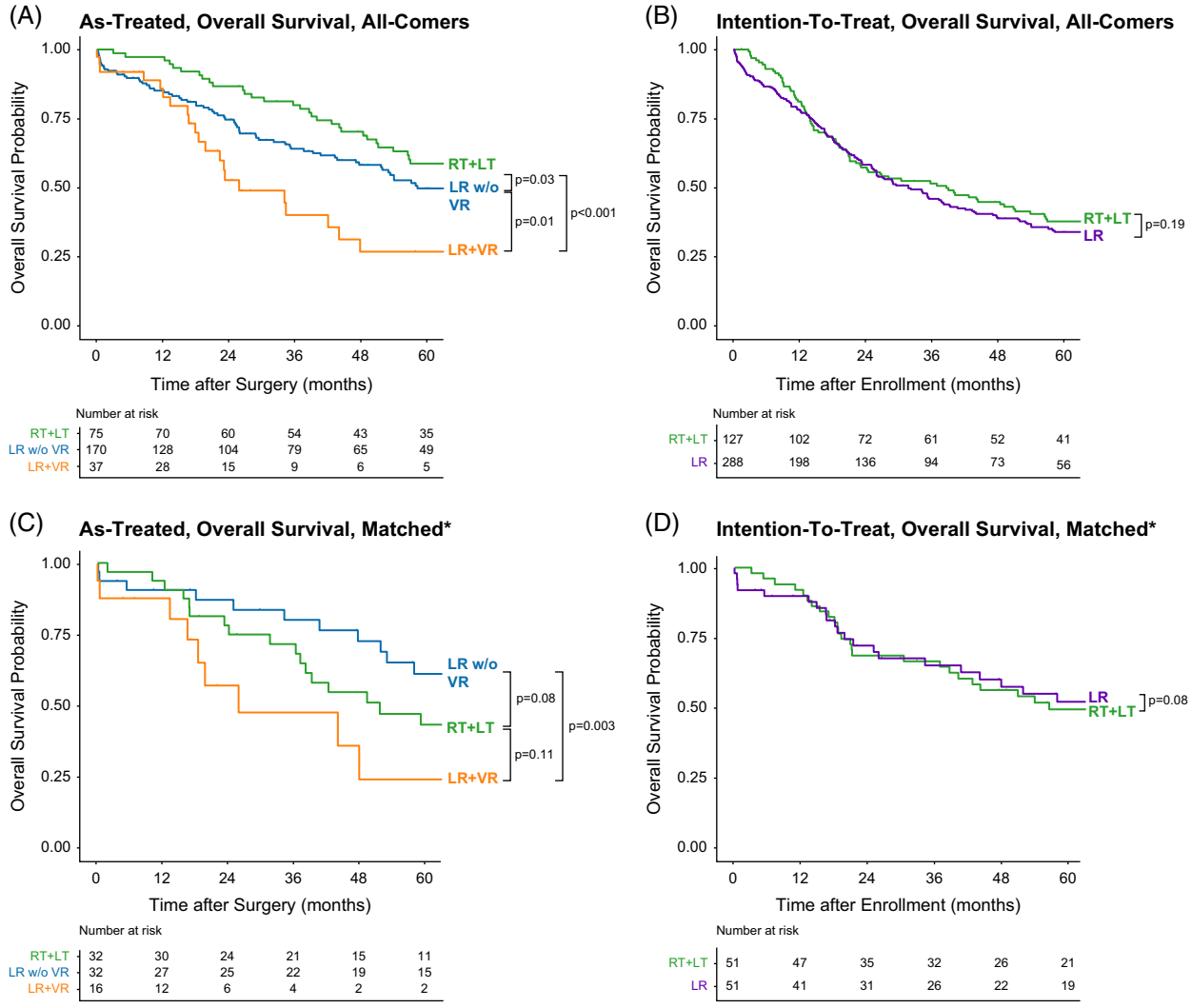
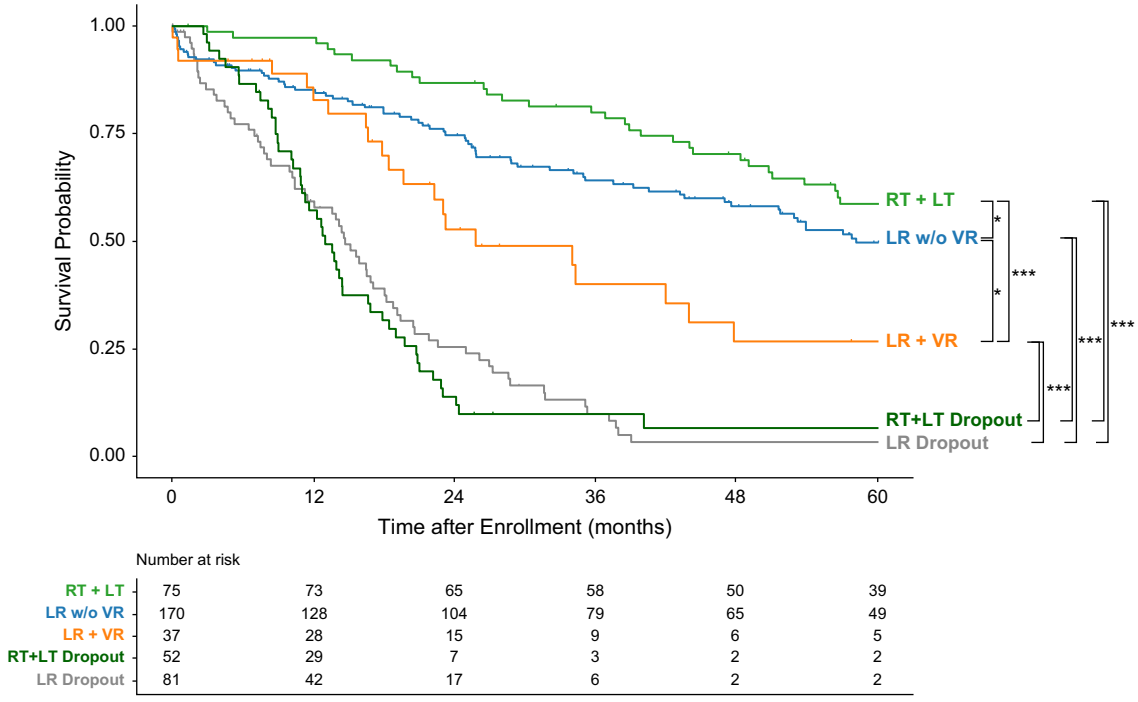


FIGURE 2 Kaplan-Meier curves estimating the OS of all patients undergoing liver transplantation following neoadjuvant chemoradiation (RT+LT), and those who underwent liver resection with vascular resection (LR+VR) and without vascular resection (LR w/o VR). (A, B) The median follow-up times were 16.3 years for RT+LT, 5.8 years for LR+VR, and 7.0 years for LR w/o VR. The median OS was superior for the RT+LT when compared with LR w/o VR ($p = 0.03$) as well as with LR+VR ($p < 0.001$). There was also a statistically significant difference in median OS favoring LR w/o VR over LR+VR ($p = 0.01$). However, in the intention-to-treat analysis, no statistically significant difference was observed between RT+LT and LR ($p = 0.19$). (C, D) In matched cohorts, median OS for the RT+LT group was comparable to LR+VR ($p = 0.11$) but tended to be inferior to LR w/o VR ($p = 0.08$) in as-treated analysis, while in the intention-to-treat analysis, there was no statistically significant difference in median OS between RT+LT and LR ($p = 0.8$). *Matching criteria for the as-treated analysis: age, sex, negative nodal status, tumor size < 3 cm, CA19-9 < 200 U/mL, and vascular involvement. Matching criteria for the intention-to-treat analysis: age, sex, negative nodal status, tumor size < 3 cm, and CA19-9 < 200 U/mL. Abbreviations: CA19-9, carbohydrate antigen 19-9; LR, liver resection; OS, overall survival; RT + LT, neoadjuvant chemoradiation and liver transplantation; VR, vascular resection.

significantly lower incidence than LR+VR ($p = 0.008$), and RT+LT showed a trend toward superiority over LR+VR ($p = 0.06$, Figure 4B).

After matching (RT+LT, n = 32; LR w/o VR, n = 32; LR+VR, n = 16), the LR w/o VR group remained the most favorable group with the highest RFS, followed by RT+LT and LR+VR. However, the difference between LR w/o VR and RT+LT was not statistically significant ($p = 0.12$), suggesting comparable outcomes between these 2 groups (Figure 4C).

Finally, when the cumulative incidence of recurrence was analyzed, accounting for competing risks and after matching, the same trend as in the all-comers cohort persisted, but with stronger statistical significance between the 3 cohorts. This time, the differences between LR w/o VR and RT+LT reached statistical significance ($p = 0.001$), with an even stronger distinction between LR w/o VR and LR+VR ($p < 0.001$), whereas the difference between RT+LT and LR+VR remained nonsignificant ($p = 0.23$, Figure 4D).



Number at risk

RT + LT	75	73	65	58	50	39
LR w/o VR	170	128	104	79	65	49
LR + VR	37	28	15	9	6	5
RT+LT Dropout	52	29	7	3	2	2
LR Dropout	81	42	17	6	2	2

FIGURE 3 Kaplan-Meier curves for OS in LR with and without VR (LR+VR vs. LR w/o VR), liver transplantation (RT+LT), and dropouts for LR versus RT+LT groups. *denotes $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Abbreviations: LR, liver resection; OS, overall survival; RT + LT, neoadjuvant chemoradiation and liver transplantation; VR, vascular resection.

OS and RFS subgroup analysis accounting for tumor size (<3 cm vs. ≥ 3 cm), resection margin (R0 vs. R1), and nodal status (N0 vs. N1) for patients undergoing LR

Further subgroup analysis was performed to evaluate OS and RFS between patients undergoing RT+LT and LR while accounting for key risk factors within the LR cohort, including tumor size (<3 cm vs. ≥ 3 cm), resection margin status (R0 vs. R1), and nodal involvement (N0 vs. N1), as shown in Figure 5. Notably, no significant differences in OS ($p = 0.37$) or RFS ($p = 0.49$) were observed between patients undergoing RT+LT (n = 75) and patients undergoing LR with negative lymph node status (N0, n = 103). A similar trend was seen for resection margin status, as median OS ($p = 0.11$) and RFS ($p = 0.15$) in patients undergoing LR with negative margins (R0, n = 166) did not differ significantly from those in the RT+LT cohort (n = 75). In addition, when stratified by tumor size (<3 cm vs. ≥ 3 cm), patients undergoing LR with smaller tumors (n = 161) showed no significant difference in OS ($p = 0.08$) or RFS ($p = 0.13$) compared with patients undergoing RT+LT (n = 75).

Furthermore, 72 (43%) patients undergoing LR and 54 (72%) patients undergoing RT+LT met previously established benchmark criteria for liver resection and transplantation, respectively.[21,24] The comparison of median OS between these benchmark groups revealed no statistically significant difference ($p = 0.85$, Supplemental Figure S3, http://links.lww.com/HEP/J874).

DISCUSSION

This review of 3 decades of multidisciplinary management of de novo pCCA at Mayo Clinic Rochester highlights 3 key findings: (1) For patients with resectable de novo pCCA, ITT survival after LR is comparable to that of patients with unresectable pCCA enrolled in the neoadjuvant protocol with the intent to undergo RT+LT, both delivering excellent 5-year survival rates. (2) Appropriate patient selection is crucial as those who do not undergo surgery (LT or LR) suffer from a poor prognosis. (3) OS doubles in patients who received LR+VR compared with dropouts, suggesting that LR+VR is a valuable treatment option for patients not eligible for RT+LT. This is the largest single-center experience on both LR and RT+LT for de novo pCCA, providing critical evidence to resolve a key debate: RT+LT should be reserved for unresectable cases, while, in patients who do not fulfill transplant criteria, LR+VR can be considered when high surgical expertise is available to expand the boundaries of resectability.

The surgical management of de novo pCCA remains highly complex, requiring careful patient evaluation, selection, and a strategic decision between RT+LT and LR+VR or LR w/o VR. Accurate assessment of resectability is critical, as is a clear understanding of the expected outcomes and limitations of each approach.[30] This decision is particularly consequential given the lack of crossover between these treatment strategies. RT+LT offers excellent outcomes for unresectable cases but is limited by a high dropout rate. In

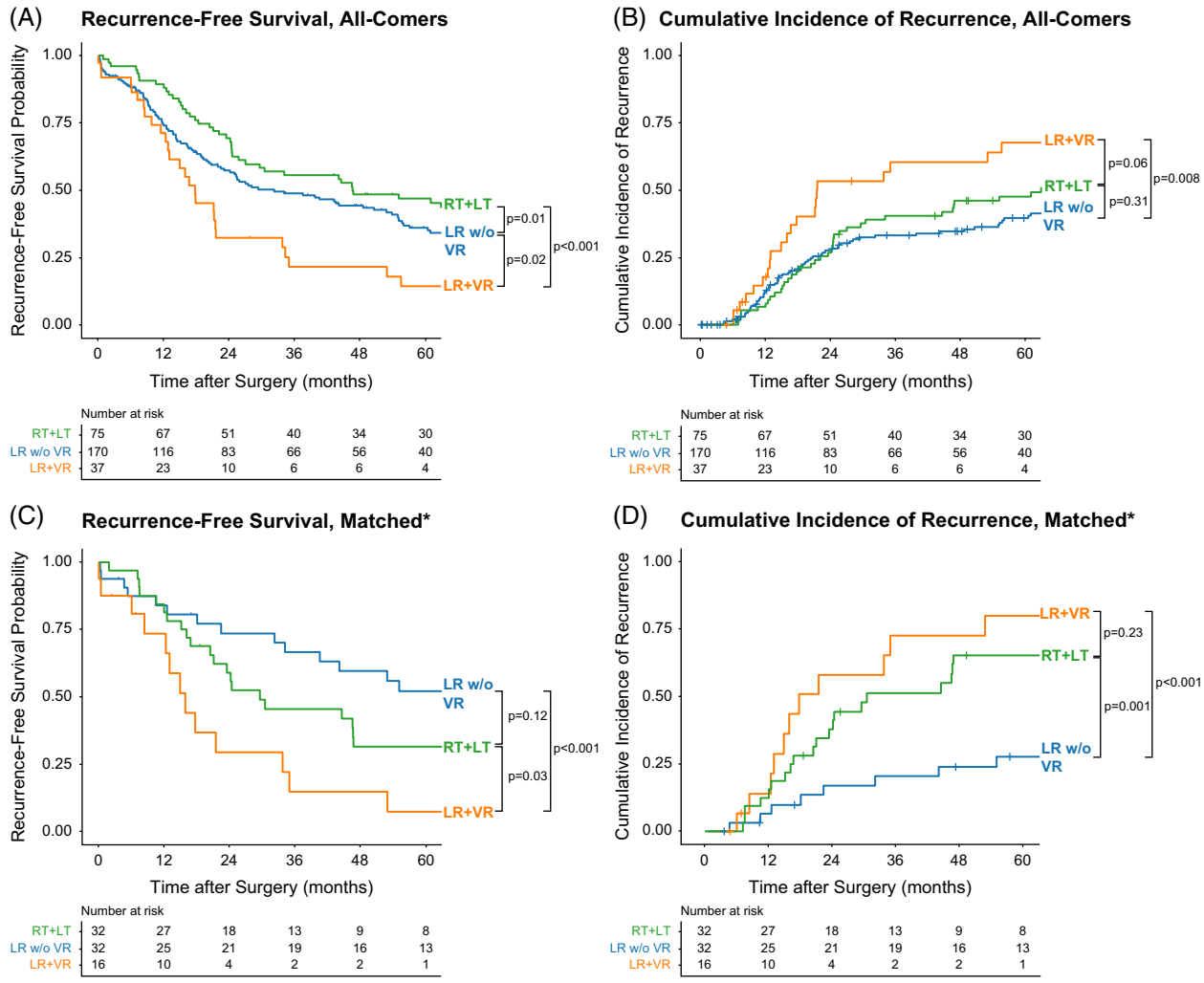


FIGURE 4 RFS and cumulative incidence of recurrence analysis incorporating competing risks for all-comers and in matched cohorts. (A, B) In the all-comers cohort, the initial RFS analysis identified RT+LT as the most favorable group, followed by LR w/o VR, while LR+VR exhibited the lowest RFS, with all pairwise comparisons being statistically significant (RT+LT vs. LR w/o VR: $p = 0.01$; RT+LT vs. LR+VR: $p < 0.001$; LR w/o VR vs. LR+VR: $p = 0.02$). When accounting for competing risk with the exclusion of non–recurrence-related events, LR w/o VR had the most favorable cumulative incidence of recurrence in the all-comers setting. The difference between LR w/o VR and RT+LT was, however, not significant ($p = 0.31$). In contrast, RT+LT showed a trend toward superiority over LR+VR in the cumulative incidence of recurrence analysis ($p = 0.06$). (C, D) After direct matching, RFS was the highest in the LR w/o VR group, followed by RT+LT and LR+VR, with these differences being statistically significant only in the RT+LT versus LR+VR ($p = 0.03$) and LR w/o VR versus LR+VR ($p < 0.001$) comparison. When adjusting for competing risk, the difference between LR w/o VR and RT+LT was, this time, statistically significant ($p = 0.001$), whereas the cumulative incidence of recurrence was comparable between RT+LT and LR+VR ($p = 0.23$). *Matching criteria for the as-treated analysis: age, sex, negative nodal status, tumor size < 3 cm, CA19-9 < 200 U/mL, and vascular involvement. For the intention-to-treat analysis: age, sex, negative nodal status, tumor size < 3 cm, and CA19-9 < 200 U/mL. Abbreviations: CA19-9, carbohydrate antigen 19-9; LR, liver resection; RFS, recurrence-free survival; RT + LT, neoadjuvant chemoradiation and liver transplantation; VR, vascular resection.

this study, 41% of patients with de novo pCCA could not proceed to LT, primarily due to distant metastases and regional lymph node involvement.[11,31] This dropout rate aligns with previously reported rates for de novo pCCA (38%) but is notably higher than the 31% observed in the mixed cohort of de novo and PSC-associated pCCA and the 15% in the PSC-only cohort.[31] As reported in prior studies, patients with PSC-associated pCCA have a more favorable prognosis and lower dropout rates, likely due to earlier detection, closer surveillance, and less aggressive tumor biology. These findings underscore the more aggressive nature of de novo pCCA and highlight the need for refined selection criteria and intensified pretransplant monitoring in this high-risk group.

While neoadjuvant chemoradiation provides early local tumor control, it is also associated with significant toxicity. Nearly all patients experience recurrent (sub) clinical cholangitis, while vascular friability frequently leads to ischemic cholangiopathy and strictures.[31] Without transplantation, these complications often progress to radiation-associated liver failure starting after about 9 months, as observed in this study. Notably, all transplant dropouts had already undergone

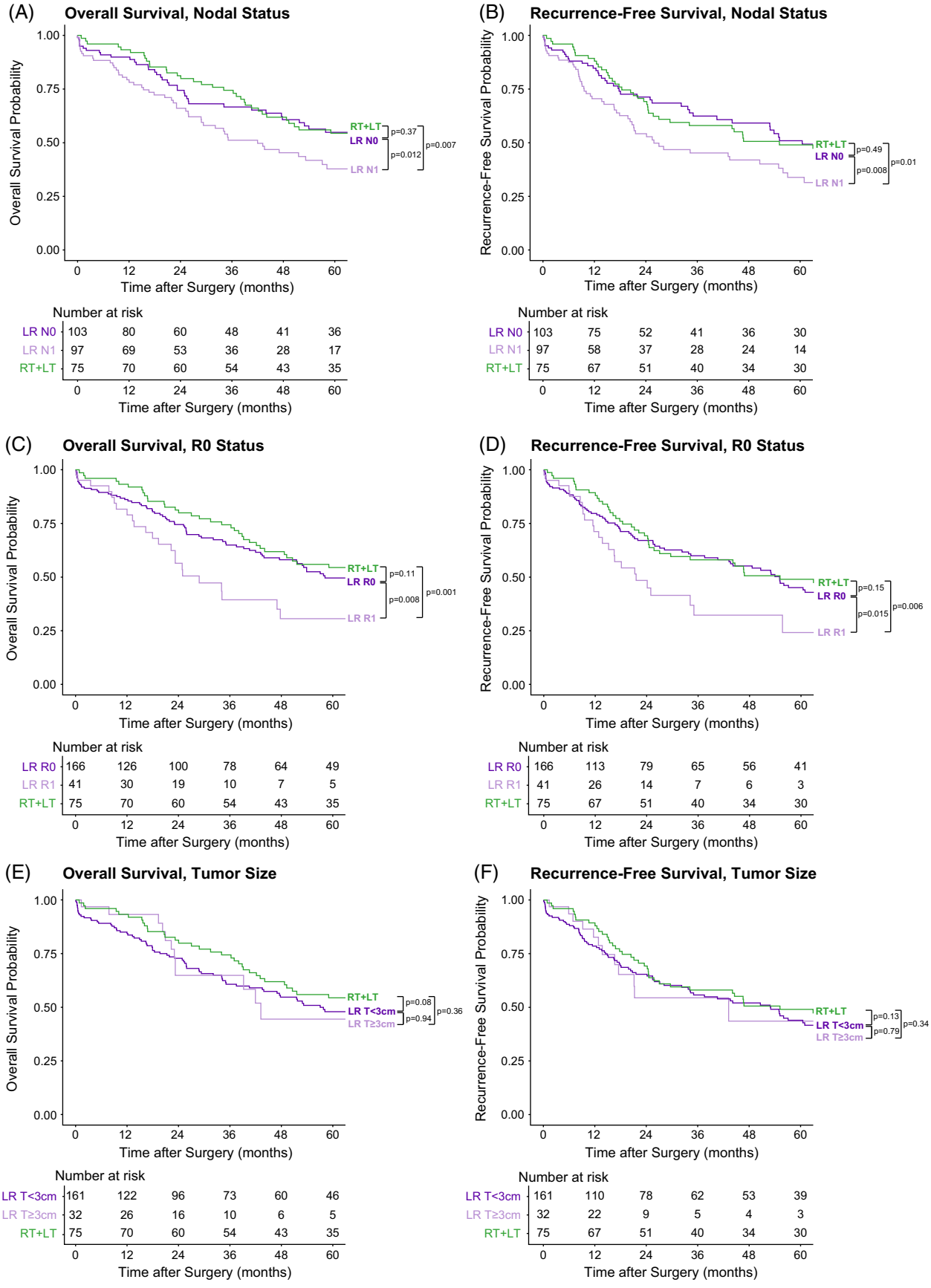

(A)
Overall Survival, Nodal Status
Overall Survival Probability
1.00
0.75
0.50
0.25
0.00
RT+LT
LR N0
LR N1
p=0.37
p=0.012
p=0.007
0
12
24
36
48
60
Time after Surgery (months)
Number at risk
LR N0 103 80 60 48 41 36
LR N1 97 69 53 36 28 17
RT+LT 75 70 60 54 43 35
Time after Surgery (months)
(B)
Recurrence-Free Survival, Nodal Status
Recurrence-Free Survival Probability
RT+LT
LR N0
LR N1
p=0.49
p=0.008
p=0.01
Time after Surgery (months)
Number at risk
LR N0 103 75 52 41 36 30
LR N1 97 58 37 28 24 14
RT+LT 75 67 51 40 34 30
Time after Surgery (months)
(C)
Overall Survival, R0 Status
Overall Survival Probability
RT+LT
LR R0
LR R1
p=0.11
p=0.008
p=0.001
Time after Surgery (months)
Number at risk
LR R0 166 126 100 78 64 49
LR R1 41 30 19 10 7 5
RT+LT 75 70 60 54 43 35
Time after Surgery (months)
(D)
Recurrence-Free Survival, R0 Status
Recurrence-Free Survival Probability
RT+LT
LR R0
LR R1
p=0.15
p=0.015
p=0.006
Time after Surgery (months)
Number at risk
LR R0 166 113 79 65 56 41
LR R1 41 26 14 7 6 3
RT+LT 75 67 51 40 34 30
Time after Surgery (months)
(E)
Overall Survival, Tumor Size
Overall Survival Probability
RT+LT
LR T<3cm
LR T≥3cm
p=0.08
p=0.94
p=0.36
Time after Surgery (months)
Number at risk
LR T<3cm 161 122 96 73 60 46
LR T≥3cm 32 26 16 10 6 5
RT+LT 75 70 60 54 43 35
Time after Surgery (months)
(F)
Recurrence-Free Survival, Tumor Size
Recurrence-Free Survival Probability
RT+LT
LR T<3cm
LR T≥3cm
p=0.13
p=0.79
p=0.34
Time after Surgery (months)
Number at risk
LR T<3cm 161 110 78 62 53 39
LR T≥3cm 32 22 9 5 4 3
RT+LT 75 67 51 40 34 30
Time after Surgery (months)

FIGURE 5 Subgroup analysis comparing OS and RFS between patients undergoing RT+LT and patients undergoing LR accounted for key risk factors within the LR cohort: (A, B) nodal involvement (N0 vs. N1), (C, D) resection status (R0 vs. R1), and (E, F) tumor size (<3 cm vs. ≥ 3 cm). No significant differences in OS or RFS were observed between patients undergoing RT+LT and patients undergoing LR with negative lymph nodes (N0), negative margins (R0), or smaller tumors (<3 cm). Abbreviations: LR, liver resection; OS, overall survival; RFS, recurrence-free survival; RT + LT, neoadjuvant chemoradiation and liver transplantation.

intensive chemoradiation and brachytherapy without the possibility of crossing over to resection. To mitigate late biliary complications, conformal hyperfractionated external beam radiation[32] and stereotactic body radiotherapy[33] have been proposed as alternatives to brachytherapy, though direct comparisons remain lacking. The high toxicity of the neoadjuvant chemoradiation protocol underscores the need for careful patient selection, with a thorough risk assessment before enrollment. Our findings suggest that patients with vascular involvement who undergo LR+VR can achieve significantly better outcomes than those who drop out during neoadjuvant therapy before LT. This underscores the importance of reducing dropout rates for de novo CCA as a key strategy to further improve patient outcomes. Several strategies could be envisioned to reduce transplant dropout rates in de novo pCCA. For instance, before initiating neoadjuvant treatment, a minimally invasive lymphadenectomy combined with peritoneal staging could help identify clinically occult lymph node metastases or minimal peritoneal disease, enabling more precise patient selection. Further advances in imaging, metabolic imaging, and circulating tumor DNA assessment might allow for better patient selection in the future.

LR allows for immediate tumor removal, avoiding the delays associated with RT+LT.[22,34] While the selection of biology is certainly key in the success of RT+LT in pCCA, timely intervention has been associated with a lower dropout rate compared with LT, which is also reflected in this study. Specifically, the LR group had a dropout rate of 28%, notably lower than the 41% observed in the RT+LT cohort, primarily due to locally advanced disease, such as vascular invasion, distant lymph node involvement, or metastases (eg, peritoneal carcinomatosis, intrahepatic spread). However, the procedure is technically demanding, particularly when vascular reconstruction is required, which increases morbidity and mortality. In addition, ensuring a sufficient future liver remnant is critical to prevent posthepatectomy liver failure.[35,36] On that note, our results illustrate that postoperative mortality remains the main challenge for LR for de novo pCCA, with a 90-day mortality of 7% in LR w/o VR and 8% in LR + VR. In our study, the impact of 90-day mortality on OS is reflected in the initial drop in survival curves, which then remain parallel to those of the transplant group. Of note, when excluding perioperative deaths in the competing risk analysis, the cumulative incidence of recurrence between RT+LT and LR w/o VR was comparable. Of interest, after matching for vascular encasement in the RT+LT cohort, its cumulative recurrence incidence closely aligned with the LR+VR group, suggesting tumor biology, rather than surgical approach, as a significant determinant of poor outcomes in the LR+VR cohort. Compared with the Nagoya center, another reference institution for pCCA resection with reported perioperative mortality rates of 1.2% for LR w/o VR and 3.6% for LR+VR and long-term survival of 50.1% and 27.0% at 5 years, respectively, our center's survival outcomes were comparable.[37] These findings support LR+VR as a viable option for patients deemed unresectable otherwise and not eligible for RT+LT, offering superior outcomes to palliative systemic therapy.

In recent years, the proposal of offering RT+LT to patients with resectable disease has also evolved, based on the premise that total hepatectomy ensures the best possible margin, thereby addressing the challenge of achieving tumor-free margins, particularly in cases with extensive hilar involvement.[21,22] To date, 2 per-protocol retrospective analyses have demonstrated that patients with resectable pCCA meeting transplant criteria (<3 cm, lymph node-negative) had significantly worse survival after LR compared with those undergoing RT +LT for similarly staged but unresectable disease.[21,22] Notably, these 2 studies included both patients with de novo pCCA and patients with PSC-associated pCCA, with the majority of transplanted cases being PSC-associated pCCA. In our matched cohort of patients with de novo pCCA, this survival advantage of RT+LT over LR was not observed in the AT analysis. Given the impact of protocol dropouts, the ITT analysis provides a more accurate assessment of treatment effectiveness. Here, our matched analysis also found no significant survival benefit for RT+LT over LR. Our findings align with the open-label, randomized *TRANSPHIL* (NCT02232932) trial, which compared neoadjuvant chemoradiation followed by liver transplantation ($n=20$) with liver resection ($n=20$) for resectable pCCA. Although both arms had a 5-year OS of only 30%, the LT group had a 50% dropout rate among initially resectable patients, leading to the trial's early termination for ethical concerns. These findings reinforce that RT+LT should be reserved for unresectable cases, especially in light of the intrinsic risks of dropout in de novo pCCA.

While our findings indicate that resectable de novo pCCA should undergo LR, they also suggest a rationale for broadening LT criteria. Our analysis of high-risk subgroups—based on resection margin status, lymph node involvement, and tumor size—found no significant survival difference between recipients of LT with tumors <3 cm, LR for tumors ≥ 3 cm, and tumors <3 cm. This challenges the arbitrary 3 cm cutoff in the transplant

criteria. Given that first-line systemic therapy for biliary tract cancer still yields poor outcomes, with a median survival of just over 12 months,[38] this might offer additional treatment options for this patient group.

This study has several limitations. Its retrospective, single-center design may introduce selection bias and limit generalizability. Despite balanced matching, the small cohort of patients undergoing LR + VR increases the risk of type II error. Another inherent limitation of this study is the extended observation period of 30 years, during which time-related biases may have influenced outcomes. These include evolving paradigms in oncologic treatment, advancements in surgical techniques, and improvements in perioperative care, including the adoption of more refined imaging modalities. Accordingly, within our resection cohort of patients with de novo pCCA, we observed a temporal increase in the use of systemic therapy in both neoadjuvant and adjuvant settings, particularly in recent years, as illustrated in Supplemental Figure S4, http://links.lww.com/HEP/J874. Furthermore, with advancements in oncologic treatment, additional tools such as next-generation sequencing are being increasingly explored for their role in guiding personalized therapeutic strategies and refining patient selection. In this context, we assessed the use of next-generation sequencing within our resection cohort of patients with de novo pCCA (Supplemental Figure S5, http://links.lww.com/HEP/J874) and analyzed the distribution of adjuvant treatment regimens across different surgical eras (Supplemental Table S6, http://links.lww.com/HEP/J874). Beyond temporal biases, the study also faced methodological complexities related to the definition of ITT for the LR cohort. Therefore, we developed 2 distinct ITT approaches: one that classified patients with distant metastases, diagnosed during exploratory laparoscopy, as dropouts from the LR cohort and another that excluded these patients entirely. This differentiation was necessary, as exploratory laparoscopy can also serve as a diagnostic staging tool, prompting questions about the applicability of the traditional ITT framework in this context. Consequently, one could argue that including patients with distant metastases in the ITT analysis of patients undergoing LR does not fully align with the concept of treatment intent, as their exclusion from surgery is based on intraoperative findings rather than initial treatment strategy. Accordingly, we illustrated both options—one in the main section and the other in the supplemental section of this manuscript, http://links.lww.com/HEP/J874.

CONCLUSIONS

In conclusion, this study provides critical insights into the optimal surgical approach for de novo pCCA, emphasizing that treatment decisions should be guided by tumor biology, resectability, and surgical expertise. While current indications and protocols deliver excellent 5-year survival rates for RT+LT and LR, key challenges remain in LT dropout rates and perioperative mortality after LR. RT+LT offers the best outcomes for unresectable patients, but its benefits over LR in patients with resectable de novo pCCA were not observed in our analysis. LR+VR can expand resectability and improve survival when performed in a high-volume center to minimize perioperative mortality. Given these findings, RT+LT should be reserved for unresectable cases, while LR remains the preferred approach for resectable de novo pCCA. The decision for borderline cases remains particularly challenging and must consider tumor extent, patient comorbidities, and donor organ availability, and should be performed in specialized high-volume multidisciplinary centers with dedicated transplant and hepatobiliary expertise. Ongoing advances in imaging, minimally invasive surgery, and circulating tumor DNA might allow improved baseline assessment of these patients. As multimodal treatment strategies continue to evolve, further prospective studies are needed to refine treatment algorithms and optimize outcomes for this challenging malignancy.

AUTHOR CONTRIBUTIONS

Yawen Dong: conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft, and writing—review and editing. Zhihao Li: conceptualization, data curation, formal analysis, investigation, methodology, writing— original draft, and writing—review and editing. Vanja Podrascanin: conceptualization, formal analysis, investigation, and writing—review and editing. John E. Eaton: conceptualization, formal analysis, investigation, and writing—review and editing. Sumera I. Ilyas: conceptualization, formal analysis, investigation, supervision, and writing—review and editing. Gregory J. Gores: conceptualization, data curation, formal analysis, investigation, supervision, and writing—review and editing. Susanne G. Warner: conceptualization, investigation, and writing—review and editing. Tayyab S. Diwan: conceptualization, investigation, and writing—review and editing. David M. Nagorney: conceptualization, data curation, investigation, supervision, and writing—review and editing. Julie K. Heimbach: data curation, investigation, supervision, and writing—review and editing. Rory L. Smoot: data curation, investigation, supervision, and writing—review and editing. Timucin Taner: data curation, investigation, supervision, and writing—review and editing. Patrick P. Starlinger: conceptualization, data curation, formal analysis, investigation, methodology, supervision, writing—original draft, and writing—review and editing.

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CONFLICTS OF INTEREST

Sumera I. Ilyas consults for AstraZeneca. The remaining authors have no conflicts to report.

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